

COMPANION MODE SPEC

Accessibility.

Three companion modes.

The standard system already meets WCAG AA at every text combination and AAA on the primary graphite scale. Accessibility is the baseline, not a mode. These three companion switches — Larger text, High contrast, Reduced motion — extend that baseline for users who need more than compliance.

They are independent toggles that combine freely with each other and with dark mode. Sixteen possible states; four of them need explicit spec.

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• 01 – POSITION

Four principles.

Companion modes are additive infrastructure — they layer on top of a system already designed for accessibility, not compensating for a system that isn't.

01

Baseline first.

The standard light and dark systems already meet WCAG AA at every combination and AAA on the primary graphite scale. Cobalt at 8.4:1 AAA in light, 7.8:1 AAA in dark. Companion modes are not fixes for a broken baseline — they are extensions for users whose needs exceed compliance.

02

Three independent toggles.

Larger text, High contrast, and Reduced motion are separate switches. Not one bundled "accessibility mode." A user with presbyopia may want larger text but no contrast change. A user with a vestibular disorder wants reduced motion regardless of text size. Bundling them forces false choices.

03

Honor system preferences first.

OS-level preferences — `prefers-reduced-motion`, `prefers-contrast`, browser text zoom — take effect automatically on first paint. Users who set these at the system level should not have to re-enable them per site. Manual toggles override system pref for site-specific choice.

04

Persist explicit choice.

Once a user picks manually, that choice persists in `localStorage` across sessions. System preference is the default; explicit toggle wins. This matters because users adjust their setup for the specific task — a designer might reduce motion when reading long-form case studies but not when browsing visual work.

Larger text.

Serves users with low vision, presbyopia, small-screen contexts, or reading fatigue. Body-oriented sizes scale by 1.2×; display type stays fixed because it is already large.

The core insight: users who need larger text need it for *reading*. Body 16 to body 19 is a real gain. Hero 72 to hero 86 is invisible — the display was already large enough. Scaling everything uniformly wastes vertical space and pushes content off the fold without benefit.

Scale mapping

ELEMENT	NORMAL	LARGER (1.2×)	CHANGE
Body — floor	16 / 26	19 / 30	scales
UI label	14 / 20	17 / 24	scales
Section deck	18 / 28	22 / 32	scales
Subheading (Geist)	20 / 28	24 / 32	scales
Card title	24 / 30	28 / 34	scales
Section heading	28 / 34	28 / 34	fixed
Case study cover	46 / 50	46 / 50	fixed
Homepage hero	72 / 75	72 / 75	fixed
Metric callout	40 / 40	40 / 40	fixed
Eyebrow · 11	11 / 15	13 / 18	scales (1.15×)
Caption · 12	12 / 18	14 / 20	scales (1.15×)
Wordmark (nav)	22	22	fixed · identity

Small labels (eyebrows, captions) scale at 1.15× rather than 1.2× to avoid overtaking the sizes they annotate — an eyebrow that grows from 11 to 13 stays legible without becoming a heading of its own.

Before & after

High contrast.

Serves users with low-vision conditions, some color-blindness edge cases, and glare-heavy environments. The graphite scale shifts up one stop; structural borders thicken; subtle washes are removed.

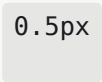
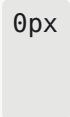
The system already runs AAA on primary text. High contrast mode targets *secondary* text — the 70 and 50 stops that are AA-large-only — by shifting them up to their next-darker sibling. Every visible text element becomes AAA at any size.

Token shift — light mode

ROLE	NORMAL	HIGH CONTRAST	VERIFIED
Body text (gr-85)	#3B3B3F · 10.5:1	#1A1A1F · 16.3:1	AAA
Secondary (gr-70)	#5D5D60 · 6.2:1	#3B3B3F · 10.5:1	AAA
Meta (gr-50)	#89898A · 3.3:1	#5D5D60 · 6.2:1	AAA
Disabled (gr-30)	#B5B5B6 · non-text	#89898A · 3.3:1	AA large
Borders (stone)	#DEDED6	#B5B5B6	stronger
Dividers (gr-18)	#D0D0CF	#89898A	stronger
Cobalt	#1F38A8 · 8.4:1	#1F38A8 · 8.4:1	unchanged

Cobalt does not shift. At 8.4:1 it is already AAA and reads as decisive against Snow. Changing it would break brand recognition without accessibility gain.

Structural changes beyond color

- Card borders thicken  to . Borders become load-bearing, not decorative.
- Focus rings thicken from  to . The current 2px reads clearly at AA; 3px reads unmistakably at any vision level.
- Cobalt-wash backgrounds (pills, callout surfaces) drop to  width — the wash becomes transparent and pills rely on their cobalt text + 1px cobalt border for definition.
- Underline treatment on links becomes always-on (removes hover-only underlines).

• 04 — REDUCED MOTION

Reduced motion.

Serves users with vestibular disorders, ADHD, migraine sensitivity, and anyone doing focused work who finds movement distracting. Every transition collapses to instant; nothing else changes.

What zeros

Every duration in the system is expressed as either `var(--dur-micro)` (200ms) or `var(--dur-page)` (350ms). Reduced motion overrides both to `0ms`. All hover states, page transitions, scroll reveals, and micro-interactions become instant.

```
@media (prefers-reduced-motion: reduce) {  
  :root {  
    --dur-micro: 0ms;  
    --dur-page: 0ms;  
  }  
}
```

What stays

- Easing curves. `var(--ease)` still applies — at 0ms it is invisible, but keeps the code path consistent between modes.
- Sequencing logic. Elements still fade in on scroll — just instantly. The *appearance* logic is preserved; only the animation is removed.
- Focus rings. Reduced motion never affects focus visibility. If anything, focus rings become more important because users cannot rely on animated confirmation of state changes.
- Loading indicators. A spinner or progress state is essential UX information, not motion decoration. Loaders remain animated even in reduced-motion mode. If a loader could be replaced with a static "Loading..." label, use the label instead.

BOUNDARY CASE — AUTO-PLAYING VIDEO

Case study cover videos, background loops, and marquee-style animations must not auto-play in reduced motion. Replace with a static poster frame and a manual play button. This is a WCAG 2.1 Success Criterion (2.3.3) and non-negotiable.

- 05 – COMBINED STATES

When modes combine.

Four independent toggles yield sixteen possible states. Twelve of them need no special handling — CSS variables cascade correctly. Four combinations need explicit spec.

The four toggles

ATTRIBUTE	VALUES	SOURCE
data-theme	"light" "dark"	manual / prefers-color-scheme
data-text	"normal" "larger"	manual / browser zoom
data-contrast	"normal" "high"	manual / prefers-contrast
data-motion	"normal" "reduced"	manual / prefers-reduced-motion

Combinations requiring explicit rules

STATE	BEHAVIOR
Dark + High contrast	gr-85 → gr-100 (in dark), Stone dark thickens to 1px, cobalt-wash drops to transparent. Cobalt Lifted stays at #8FA2EE.
Dark + Larger text	Standard body 19 on dark bg = same 13.4:1 as body 16 (color is what matters, size doesn't affect ratio). Verify only that display sizes remain visually balanced when body text grows.
High contrast + Larger text	Compounds: gr-85 becomes gr-100 (16.3:1) AND body 16 becomes body 19. Text becomes maximally legible. No conflict.
All three enabled	Maximum accessibility state. Larger text + high contrast + reduced motion. Design must remain functional here — this is the state to test against, not an edge case.

The other twelve — no special handling

The remaining twelve states inherit correctly because CSS custom properties cascade through the token layer. When larger text is enabled in dark mode, the type sizes swap but colors stay in the dark scale. When reduced motion is enabled with high contrast, motion zeros but colors shift independently. Each

• 06 — IMPLEMENTATION

Implementation.

Additions to `tokens.css` and one React hook that manages all four preferences.

tokens.css — companion mode additions

```
/* System pref defaults – apply automatically unless overridden */
@media (prefers-reduced-motion: reduce) {
  :root:not([data-motion="normal"]) {
    --dur-micro: 0ms;
    --dur-page: 0ms;
  }
}

@media (prefers-contrast: more) {
  :root:not([data-contrast="normal"]) {
    --gr-85: #1A1A1F;
    --gr-70: #3B3B3F;
    --gr-50: #5D5D60;
    --stone: #B5B5B6;
    --gr-18: #89898A;
  }
}

/* Manual overrides */
[data-motion="reduced"] {
```

React hook — usePreferences

```
const KEYS = { theme: 'theme', text: 'text-size', contrast: 'contrast', motion: 'motion'
};

export function usePreferences() {
  const [prefs, setPrefs] = useState(() => {
    if (typeof window === 'undefined') return defaults;
    const mq = window.matchMedia;
    return {
      theme:    localStorage.getItem(KEYS.theme)    ?? (mq('(prefers-color-scheme:
dark)').matches ? 'dark' : 'light'),
      text:     localStorage.getItem(KEYS.text)     ?? 'normal',
      contrast: localStorage.getItem(KEYS.contrast) ?? (mq('(prefers-contrast:
```

• 07 – TOGGLE UI

The accessibility menu.

One entry point, four toggles. Not scattered across the interface. Not hidden three levels deep. One button, one panel, four choices.

Entry point

- Icon:  (stick figure with arms out). Standard convention — Apple, Google, and every major OS use variations of this icon. Alternate:  if the figure icon reads awkwardly in a portfolio context.
- Placement: right side of desktop nav, between the theme toggle and the CTA. Bottom of the mobile drawer, above the theme toggle.
- Size: 20px icon, `--gr-70` color. Same visual weight as a nav link.
- ARIA label: `aria-label="Accessibility settings"`. Never "A11y" or icon-only in screen reader context.
- Click behavior: opens a small panel anchored to the button, dismissible via Escape, click-outside, or a close button. Not a modal — modals interrupt reading flow.

Panel contents

TOGGLE	LABEL	ARIA LABEL
Theme	Dark mode	Switch to dark mode
Text	Larger text	Enable larger text
Contrast	High contrast	Enable high contrast
Motion	Reduce motion	Reduce motion and animation

Each row is a labeled switch, not a checkbox — switches communicate "on/off state" more clearly. Below the four rows, a `Reset to system preferences` text link that clears all four localStorage keys and returns to OS-level defaults.

• 08 – INVARIANTS

What does not change